POPULATION HEALTH DATA SCIENCE: ALGORITHMIC EPIDEMIOLOGY

Decoupling Disease Detection from Healthcare Access

Evaluating Proxy Label Bias and Algorithmic Triage Fairness in Population Screening Under Finite Testing Capacity

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01 / THE CORE QUESTION

The Proxy Label Trap

Predictive screening algorithms frequently train on administrative billing codes or self-reported history. These endpoints implicitly measure healthcare utilization rather than pure pathology, penalizing undiagnosed, marginalized patients who rarely consult physicians.

N = 12,459
CDC NHANES participants curated across pre-pandemic and post-pandemic cycles.

02 / STUDY DESIGN

Temporal Validation Protocol

Development cohort: 2017 to March 2020 (N = 7,761). Holdout evaluation cohort: August 2021 to August 2023 (N = 4,698 strictly undiagnosed adults with zero temporal overlap). Model A trains on Diagnosis Only; Model B trains on Biomarker Augmented labels (Diagnosis OR HbA1c ≥ 6.5%).

03 / DATA ENGINEERING

Lakehouse & Data Contracts

Built an embedded DuckDB OLAP engine with zero-copy Parquet storage. Enforced strict Pandera data quality contracts to trap schema anomalies:

0% Male Loss
Corrected the pregnancy test trap, avoiding silent exclusion of 85% of adult males.

Target leakage controlled: Predictors strictly restricted to non-invasive demographics (age, sex, BMI, waist circumference).

04 / COMPLEX SURVEY DESIGN

Taylor Series Variance

Implemented Taylor Series Linearization across strata and PSUs (WTPH2YR weights) to generate authentic 95% national confidence intervals.

05 / EMPIRICAL FINDINGS

The 61-Patient Proof

At a realistic 20% screening capacity (N = 939 slots for 4,698 adults), Model B swaps 61 borderline patients into the screening queue:

3.0x Precision
Model B detected 3 hidden diabetes cases in swap group vs 1 for Model A ($0 added testing cost).

+$25,500
Net monetary economic benefit per 10,000 screened via Decision Curve Analysis.

06 / MODEL ROBUSTNESS

Nonlinear Stability

Validated across Random Forest and Gradient Boosting architectures: biomarker augmentation consistently yields higher sensitivity at identical test budgets.

07 / HEALTH EQUITY AUDIT

Closing the Access Divide

Under traditional models, uninsured Americans suffer a significant sensitivity deficit. Biomarker augmentation eliminates this disparity:

+8.6% Sensitivity
Direct sensitivity dividend for uninsured Americans without collecting insurance status.

The model achieves algorithmic fairness not by enforcing demographic parity constraints, but by decoupling pathology labels from healthcare utilization history.

08 / HEALTH ECONOMICS

Decision Curve Analysis

+$25,500 Net Benefit
Net monetary benefit per 10,000 screened individuals under standard health economics decision models ($25 test, $8,500 early savings).

Decision Curve Analysis (DCA) proves Model B achieves superior net clinical benefit across decision thresholds between 2% and 15%, maximizing

population health value without requiring additional lab spending ($0 extra cost).